

Chirag Parikh

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EDUCATION

- **International Institute of Information Technology, Hyderabad** 2022 - Present
Ph.D. in Computer Science and Engineering CGPA: 9/10
 - Advisor: Prof. Ravi Kiran Sarvadevabhatla
 - Thesis Topic: Grounded Road-Event Video Understanding
 - Courses: Computer Vision, Optimization Methods, Statistical Methods in AI, Digital Image Processing
- **Birla Institute of Technology, Mesra** 2013 – 2017
Bachelor of Engineering CGPA: 7.34/10
 - UG-research: Surface defect analysis and correction in Friction Stir Welding; Advisor: Prof. Surjya K. Pal
 - Winner of MathWorks Prize (ABU Robocon 2015) and 1st Place at Vertechx 2014 (robotics), and organized workshops/competitions as part of college robotics club

PHD THESIS SUMMARY

Grounded video understanding with multimodal models, progressing from ego-centric road-event explanations to grounded and entity-coreferent video reasoning. My work spans important-entity grounding and explainable driving behavior (IDD-X), large-scale road-event VideoQA and hallucination robustness (RoadSocial), and, more recently, unified VideoLLMs that jointly model what happens, when it happens, where entities appear, and who persists across events through temporal localization, spatial grounding, and entity coreference (W4VU).

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, A=ARXIV, T=THESIS

- [C.0] **Chirag Parikh**, Lakshmipathi Balaji, Hozaira Khan, Modi Akshay, Makarand Tapaswi, Ravi Kiran Sarvadevabhatla. **W4VU: Understanding Who did What, When and Where in Videos**. [Under Review]
→ Work focusing on holistic video understanding with VideoLLMs. Introduced Coreference-aware Grounded Event Captioning (CaGEC) task which jointly requires event description, temporal localization, spatial grounding of entities, and persistent entity coreferencing across events. Contributed an end-to-end method to jointly model these capabilities, plus a dataset, benchmark and task-specific evaluation metrics.
- [C.1] **Chirag Parikh**, Deepti Rawat, Rakshitha R. T., Tathagata Ghosh, Ravi Kiran Sarvadevabhatla. **RoadSocial: A Diverse VideoQA Dataset and Benchmark for Road Event Understanding from Social Video Narratives**. In *proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025 [Paper], [Project Page]
- [C.2] **Chirag Parikh**, Siddhi Lipare, Ravi Kiran Sarvadevabhatla. **RoadTones: Tone Controllable Text Generation from Road Event Videos**. In *proc. of IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR Findings)*, 2026 [Paper], [Project Page]
- [C.3] **Chirag Parikh**, Rohit Saluja, CV Jawahar, Ravi Kiran Sarvadevabhatla. **IDD-X: A Multi-View Dataset for Ego-relative Important Object Localization and Explanation in Dense and Unstructured Traffic**. In *proc. of IEEE International Conference on Robotics and Automation (ICRA)*, 2024 [Paper], [Project Page]
- [C.4] Ravi Shankar Mishra, **Chirag Parikh**, Anbumani Subramanian, C.V.Jawahar, Ravi Kiran Sarvadevabhatla. **IDD-CRS: A Comprehensive Video Dataset for Critical Road Scenarios in Unstructured Environments**. In *proc. of IEEE Intelligent Vehicles Symposium (IV)*, 2025 [Paper]
- [A.1] **Chirag Parikh**, Ravi Shankar Mishra, Rohan Chandra, Ravi Kiran Sarvadevabhatla. **Transfer-LMR: Heavy-Tail Driving Behavior Recognition in Diverse Traffic Scenarios**. [Paper], [Code]
- [C.5] Prafful Kumar Khoba, **Chirag Parikh**, Rohit Saluja, Ravi Kiran Sarvadevabhatla, C.V. Jawahar. **A Fine-Grained Vehicle Detection (FGVD) Dataset for Unconstrained Roads**. In *proc. of Thirteenth Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*, 2022 [Paper], [Code]
- [P.1] Surjya Kanta Pal, Aaquib Reza Khan, Ravi Ranjan, **Chirag Parikh**, Srikanta Pal, Debashish Chakravarty, Abhik Maiti. **Real-Time Surface Defect Analysis and Correction in Friction Stir Welding Process by Image Processing**. IN, Patent No. 514971. File Date: 20/09/2018, Grant Date: 26/02/2024. [Grant Certificate]
- [J.1] **Chirag Parikh**, Ravi Ranjan, Aaquib Reza Khan, Rahul Jain, Raju Prasad Mahto, Debashish Chakravarty, Srikanta Pal, Surjya K. Pal. **Volumetric defect analysis in friction stir welding based on three dimensional reconstructed images**. *Journal of Manufacturing Processes, Elsevier*, 2017 [Paper]
- [J.2] Ravi Ranjan, Aaquib Reza Khan, **Chirag Parikh**, Rahul Jain, Raju Prasad Mahto, Srikanta Pal, Surjya K. Pal, Debashish Chakravarty. **Classification and identification of surface defects in friction stir welding: An image processing approach**. *Journal of Manufacturing Processes, Elsevier*, 2016 [Paper]

EXPERIENCE

- **IIIT-H, Center for Visual Information Technology (CVIT) Lab** [🌐] 2022 - Present
PhD Researcher
 - Research in video question answering, driver behavior modeling and explanation, and road scene understanding.
 - Contributed to papers in CVPR 2025 and 2026, ICRA 2024 on road event understanding and driving behavior explanation in dense and unstructured traffic.
- **The Hi-Tech Robotic Systemz Ltd.** [🌐] 2019 - 2022
Senior Research Engineer (ADAS)
 - Designed HydraNet for Driver State Monitoring (DMS) with real-time inference on embedded platforms. Obtained >95% accuracy across multiple subtasks in DMS (eye-closure, smoking, yawning, phone activity).
 - CNNs optimization, custom datasets curation, and model deployment models on resource-constrained ARM devices.
- **FarmGuide** [🌐] 2017 - 2019
Computer Vision Engineer (Remote Sensing)
 - Developed CV + geospatial models for cropland detection from satellite imagery.
 - Built cloud-scalable pipelines using AWS (EC2, Lambda, S3).
 - Applied deep edge detection (HED), clustering, and time-series analysis to remote sensing data.
- **IIT Kharagpur, FSW Lab (Machine Vision)** [🌐] Summer of 2015 & 2016
Research Intern
 - Developed novel image-based methods to detect FSW defects from 3D reconstructions and time-series signals.
 - Co-invented and co-authored a patent and 2 journal papers for computer vision application in manufacturing processes.

PROJECTS

- **Differential Attention Networks for Visual Question Answering** 2023
IIIT-H Computer Vision course project advised by: Prof. Avinash Sharma
 - [Report], [Code]
- **Integrating storage manager (TASM) and query optimizer (FiGO) in Video-DBMS** 2025
IIIT-H Data Systems course project advised by: Prof. Kamal Karlapalem
 - [Report], [Code], [Video]
- **Driver Activity Recognition for Driver State Monitoring from Infrared Videos** 2021
Sr. Research Engineer at The Hi-Tech Robotic Systemz Ltd.
 - Designed a multi-task HydraNet-style architecture with BlazeFace as the shared backbone and task-specific heads for face detection, 3D 68-landmark prediction, eye state classification for drowsiness detection, and activity detection (smoking, yawning, phone usage).
 - Built end-to-end TensorFlow pipelines with TFRecord-optimized loading, custom loss functions, and advanced data augmentation strategies. Curated a 100K+ Infrared Camera Face Detection and 3D Landmark Prediction dataset using RetinaFace and 3DDFA-v2.
 - Deployed quantized TFLite models on resource-constrained ARM devices with a custom C++ inference engine, achieving real-time performance (28 FPS) and 2× speedup over MediaPipe.
 - Improved model robustness via pose-conditioned face synthesis (Conditional-BEGAN), fine-tuned HOG+SVM detectors for masked faces, and boosted eye state classification accuracy to 96% through targeted augmentations.
- **Cropland Detection from Satellite Imagery** 2018
Computer Vision Engineer at FarmGuide
 - Achieved 85% accuracy in cropland detection by segmenting farm boundaries on high-resolution satellite imagery using a transfer-learned Holistically-nested Edge Detection (HED) model. Created a farm boundaries training dataset via unsupervised Graph Cut segmentation refined through contour analysis and morphological processing.
- **Real-time Shuttlecock Tracking and Trajectory Estimation for Badminton Playing Robot** 2015
BIT Mesra Robotics Club project for ABU Robocon 2015 supported by: Prof. Arun Dayal Udai
 - [Report], [Code], [Video]

INTERESTS

Computer Vision · Video Understanding · Video-Language Modeling · Visual Grounding · Multimodal Learning · Deep Learning · Advanced Driver Assistance Systems (ADAS)

SKILLS

- **Languages:** Python, C++
- **Frameworks:** PyTorch, TensorFlow, Keras, OpenCV, dlib
- **Tools:** Git, Docker, Conda, TFDS, AWS (EC2, S3, Lambda)